

NYC Yellow Taxi: Fares, Tips & High-Fare Trips

How do time of day, day of week and pickup location shape fares and tips, and can we predict a high-fare trip using only what is known at pickup?

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Repo: github.com/vamsikrishnabethapudi02-web/nyc-taxi-project · Dashboard: NYC_Taxi_Dashboard.twbx (Tableau)

Fall 2026

47.2 million taxi trips, one row each

47.2M

raw trips in 4 monthly CSV files

11.5M

clean rows used (random 25% of every file)

19

raw columns: time, GPS, money, codes

Source

NYC Taxi & Limousine Commission trip records, via Kaggle (elemento/nyc-yellow-taxi-trip-data). January 2015 and January to March 2016.

Why this dataset

Fare, tip, time and pickup location are exactly the variables in our question. GPS, timestamps and money support maps, time patterns and classification.

Almost no missing values, but many impossible ones

Validity rule (applied in order)	Rows removed	% of rows
GPS outside NYC (incl. 0,0)	184,898	1.56%
Duration under 1 min or over 3 h	97,907	0.83%
Distance 0 or over 100 mi	13,443	0.11%
Dropoff not after pickup	12,989	0.11%
Fare 0 or over \$500	3,882	0.03%
Other 6 rules (passengers, speed, codes, duplicates)	2,645	0.02%
Clean rows kept	11,496,446	97.3%

Missing values: 1
One null in 11.8M rows, so dropna() would change nothing.

Added features
duration, speed, hour, weekday, month, weekend flag, tip % (card trips)

Four problems we hit, and what we decided

A silent bug removed all of Jan 2015

2015 says RateCodeID, 2016 says RatecodeID. One rule dropped 26% of rows. Fixed with case-insensitive names and a warning if any rule removes over 5%.

Outliers: domain limits, not IQR

Long airport trips are real. We cut only what a taxi cannot do: over 3 h, 100 mi, \$500 or 80 mph.

Cash tips are not recorded

Tip is 0 for nearly every cash trip, so all tip analysis uses credit-card trips only.

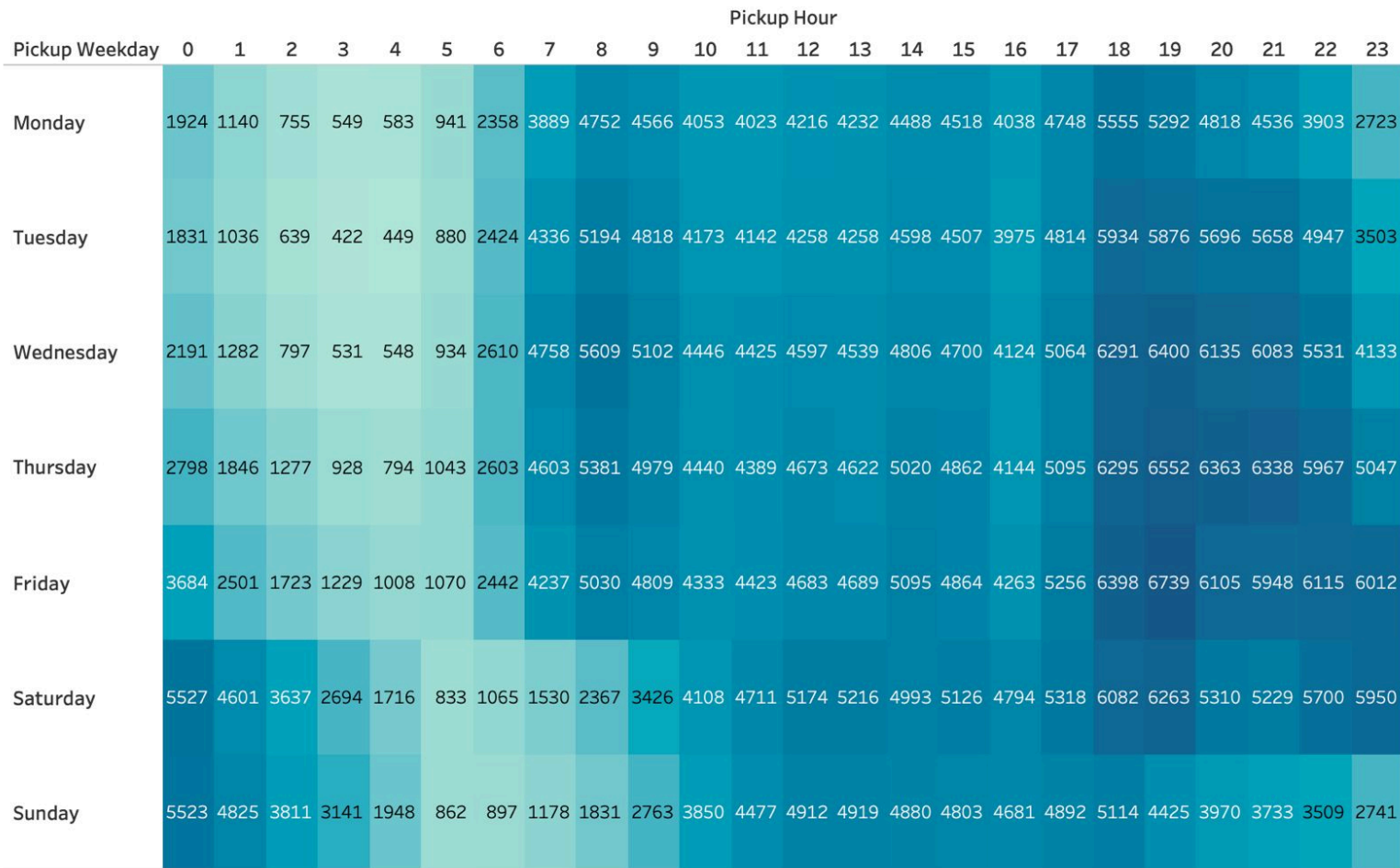
Too big for memory

47.2M rows crashed a free Colab GPU. We keep a random 25% of each file, so every month, day and hour is still covered.

Weekdays peak at 6–9pm; weekends peak after midnight

When Are NYC Yellow Taxi Pickups Busiest?

Average pickups per weekday/hour | Jan 2015 and Jan–Mar 2016



Caption
Source: NYC TLC via Kaggle. Cleaned 25% sample. Each cell shows average pickups during a one-hour period across the observed dates for that weekday. Darker colours indicate higher pickup activity.

What it shows

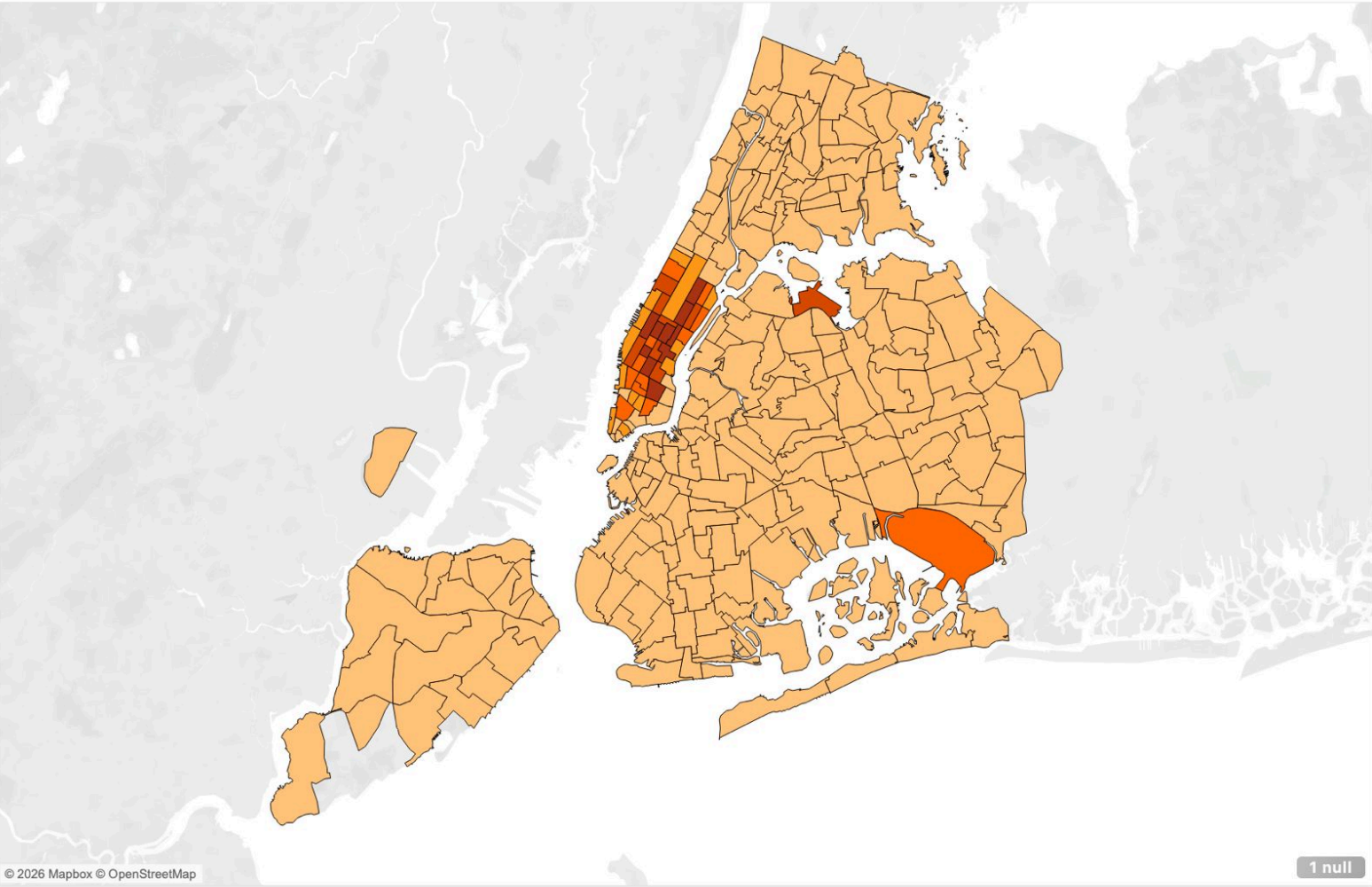
Thu and Fri 7pm are the busiest hours (6,500+ per hour). Saturday and Sunday mornings 5–8am are the quietest.

Decision
Hour and weekday interact, so the model gets hour, weekday and a weekend flag together.

Pickups concentrate in Manhattan and at the two airports

Which NYC Taxi Zones Have the Most Pickups?

Cleaned 25% sample | Jan 2015 and Jan–Mar 2016



Caption

Darker orange represents more recorded pickups. The map shows pickup activity concentrated in Manhattan, with prominent airport hotspots.

What it shows

Midtown and Lower Manhattan dominate. LaGuardia and JFK stand out as isolated hotspots in Queens.

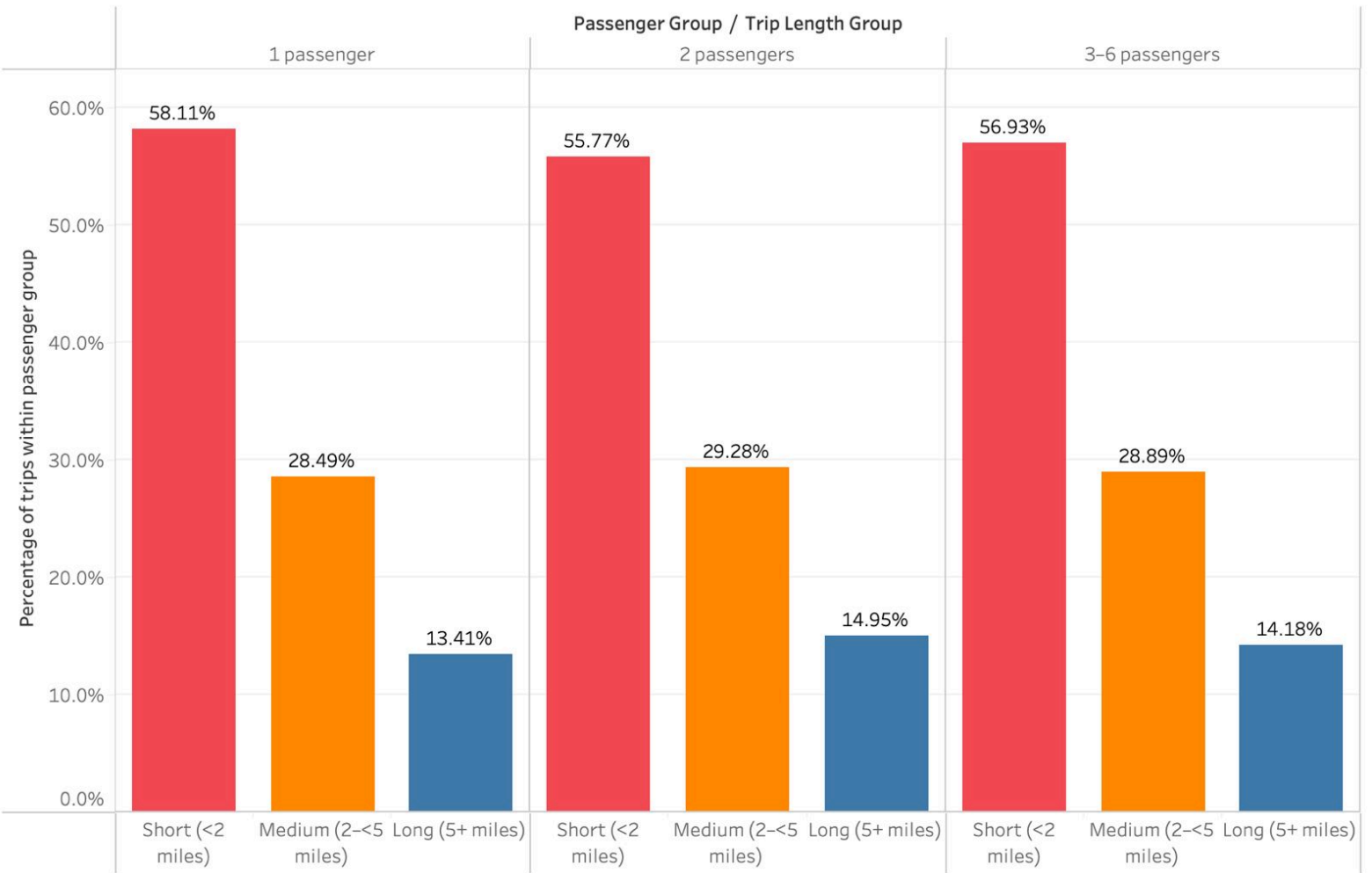
Decision

Location is uneven, so we add a pickup-area feature with a separate airport flag.

Group size barely changes trip length

Do Larger Passenger Groups Take Longer Trips?

Cleaned 25% sample / Jan 2015 and Jan–Mar 2016



Caption

Short trips dominate every passenger group. Long trips account for roughly 13–15%, so trip-length patterns are similar. The share of long trips does not increase steadily with passenger-group size.

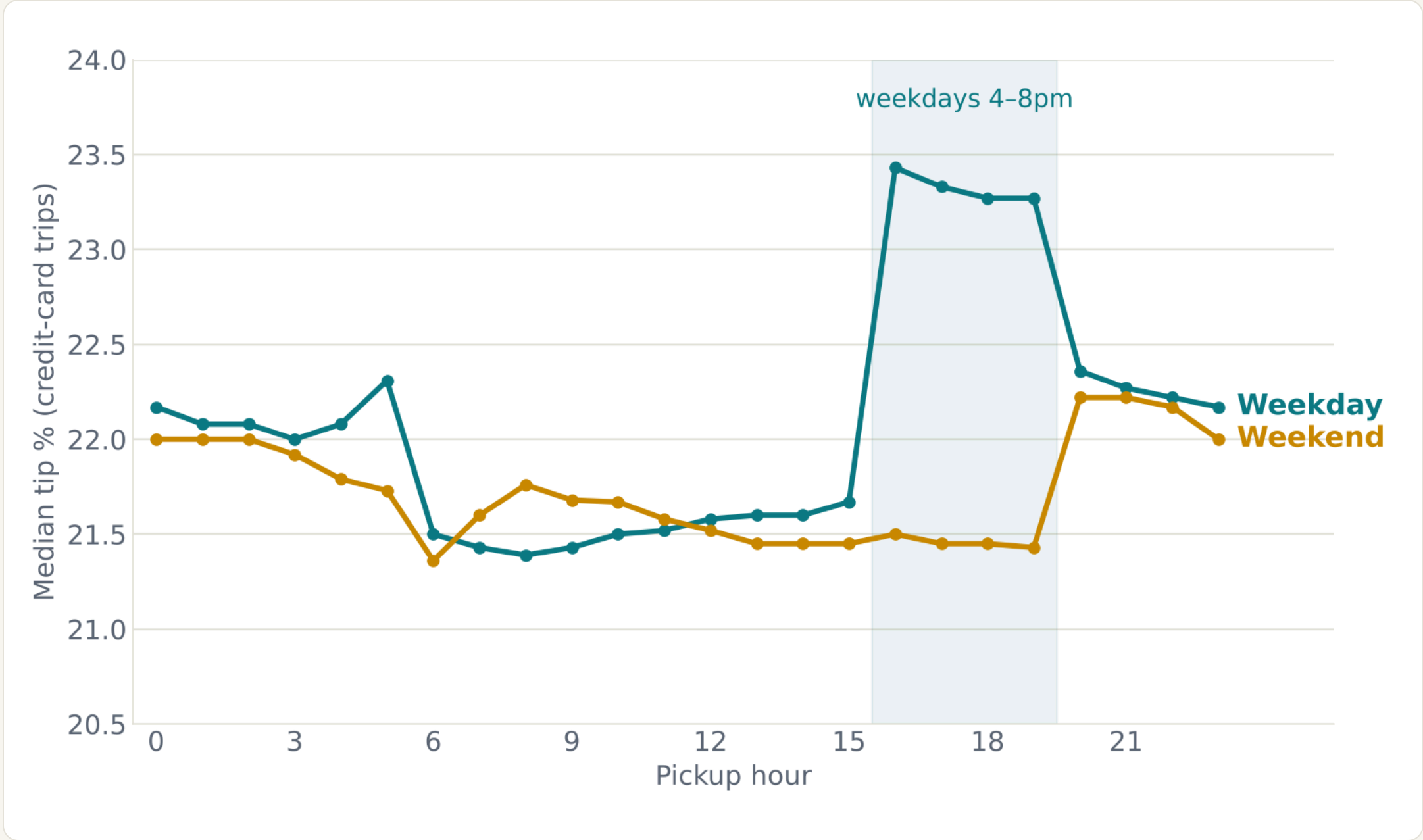
What it shows

Every group is about 57% short, 29% medium, 14% long. Long trips stay at 13–15% whatever the size.

Decision

Passenger count is a weak signal: keep it, but expect low importance.

Card tips hold near 22% all day, with a weekday 4–8pm step



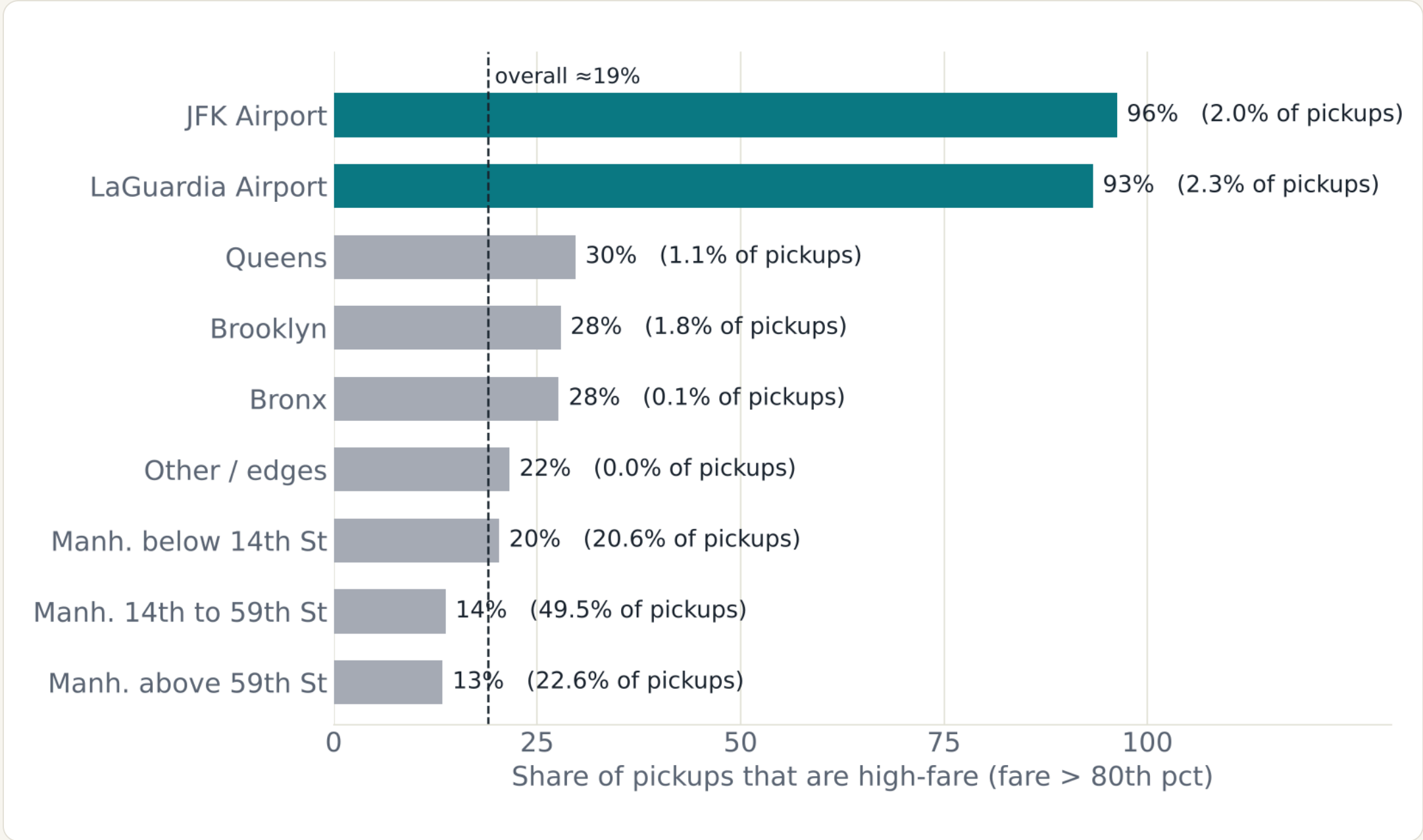
What it shows

Median tip 21.4–22.3% most hours. Weekdays 4–8pm jump to about 23.3%; weekends do not. No-tip share peaks at 3–5am (about 8%).

Decision

Time moves tips only slightly. Next: recompute tip as % of total to test the surcharge effect.

Airports: 4% of pickups, 22% of high-fare trips



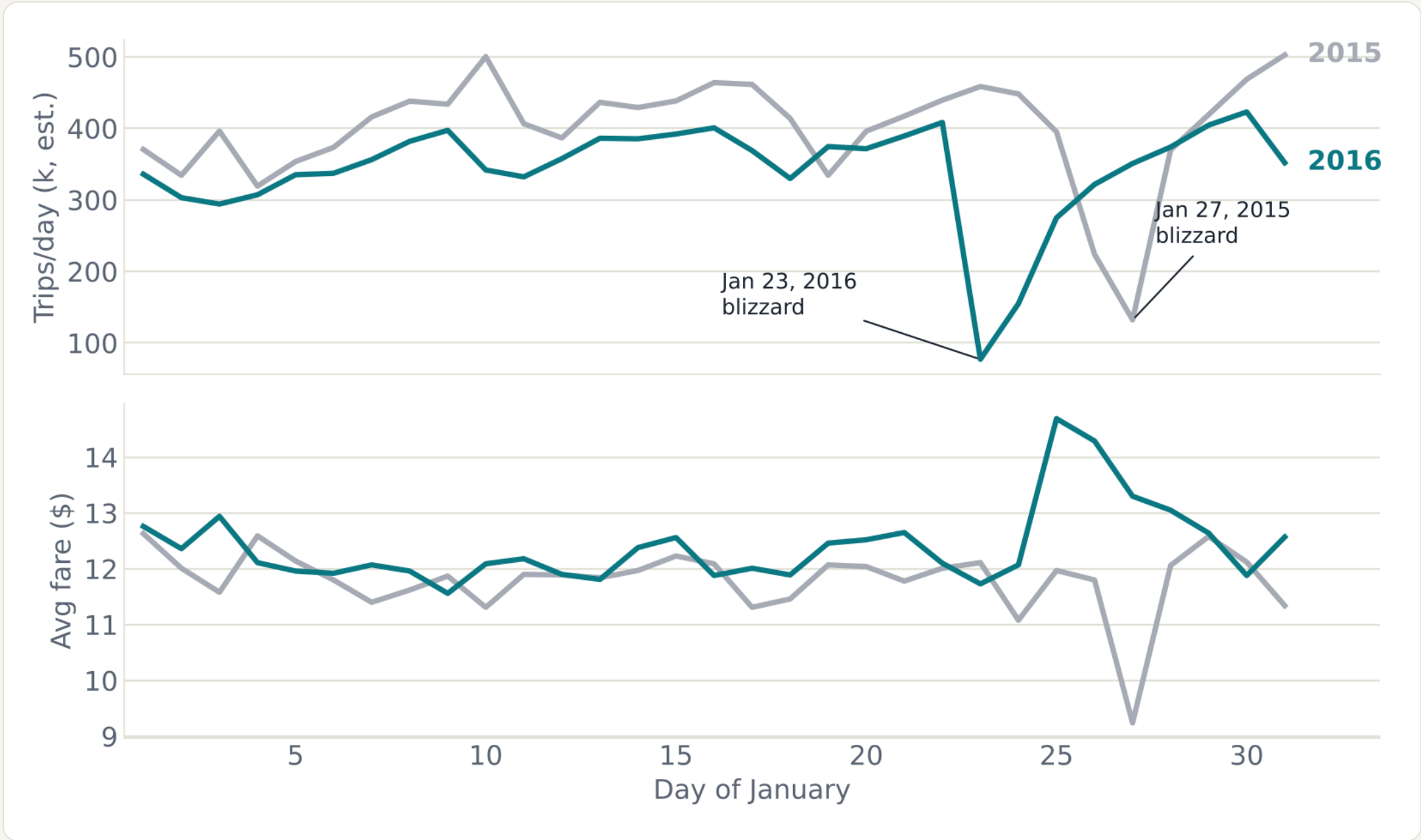
What it shows

High-fare = fare above the 80th percentile.
JFK 96% and LaGuardia 93% vs 14% in Midtown.

Decision

Pickup area is the strongest pickup-time signal. Airport trips are kept, not treated as outliers.

Jan 2016 had 14% fewer trips but 5% higher fares



What it shows

Trips per day -14.2%; average fare \$11.84 → \$12.39. Each year has one blizzard day that loses 68–78% of trips.

Decision

Year/month is a feature; storm days are real and kept. Weather data is a next step.

Six charts, six decisions for the model

Finding	Decision
1 · Peaks differ by weekday	Use hour, weekday and weekend flag together
2 · Manhattan + airport hotspots	Add pickup area and an airport flag
3 · Group size ≈ no effect	Keep passenger count, expect low importance
4 · Tips stable near 22%	Tips only on card trips; test the surcharge effect
5 · Airports 93–96% high-fare	Pickup area is the key feature; keep airport trips
6 · Volume –14%, storm days	Add year/month; keep storm days; weather later

Classify: will this trip be high-fare? (known at pickup only)

TARGET

Fare above the 80th percentile

About 19% positive, roughly 1 : 4 imbalance. Fares are skewed, so a percentile beats a round dollar cut-off.

HANDLING IMBALANCE

Class weights vs SMOTE

"Never high-fare" would be 81% accurate and useless, so we judge by F1 and PR-AUC, not accuracy.

MODELS

Logistic regression → tree models

Baseline first, then random forest or gradient boosting. Test on March 2016.

Inputs: hour, weekday, weekend, month/year, pickup area, passengers, vendor

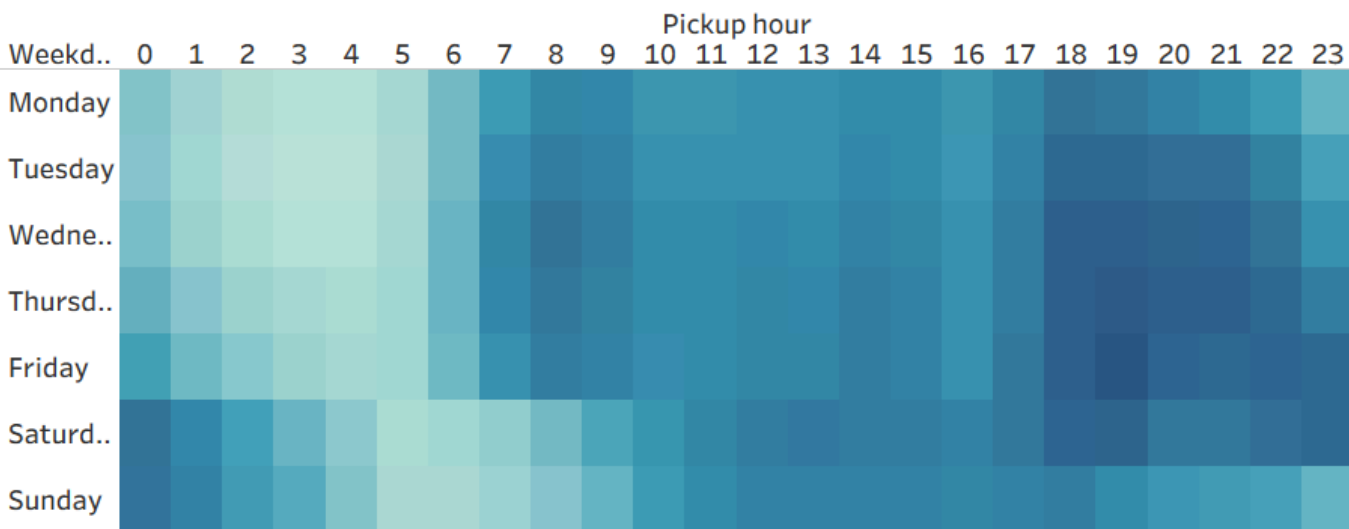
Never inputs (leakage): distance, duration, dropoff, tip, total amount

Busy times and expensive trips are not the same thing

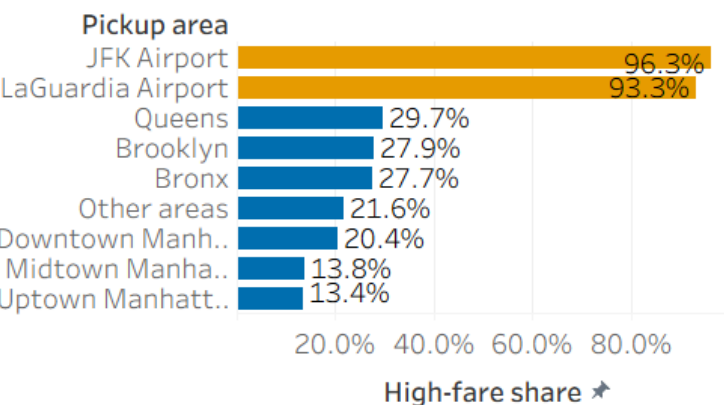
NYC Yellow Taxi: when, where, and how expensive

11.5M clean trips (random 25% of Jan 2015 and Jan-Mar 2016) | Airports: 4.3% of pickups, 21.6% of high-fare trips | High-fare = far..

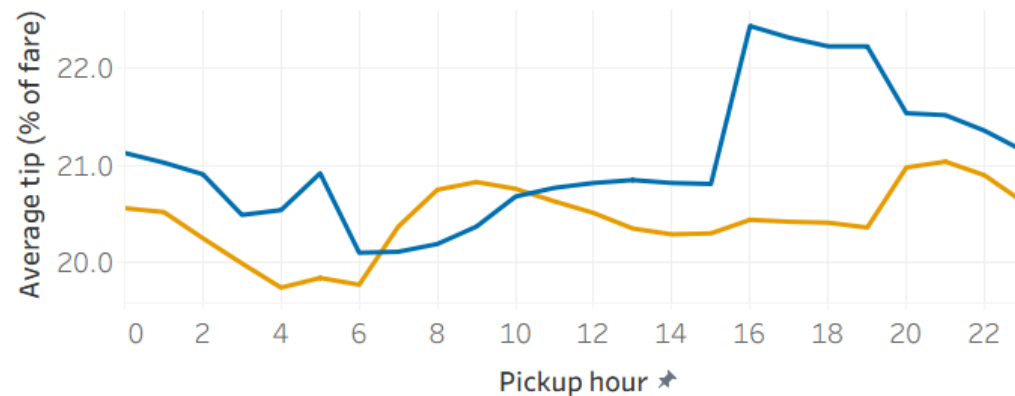
Weekdays peak 6-9 pm; weekends peak after midnight



Airport pickups are over 93% high-fare; Midtown and Uptown only 13-14%

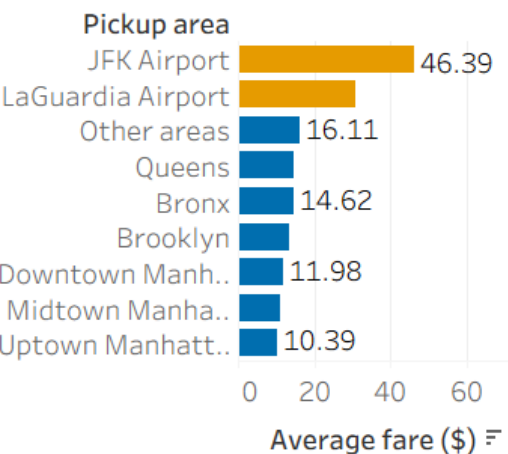


Card tips stay near 20-22% all day; the weekday 4-8 pm bump matches the rush-hour surge

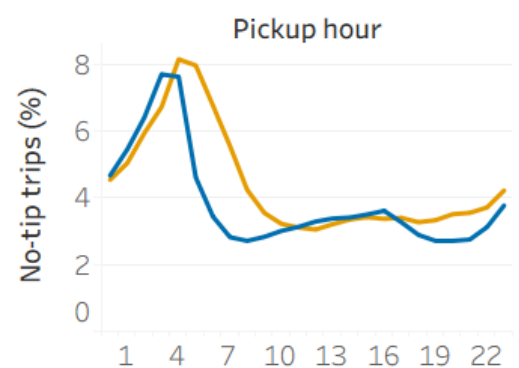


Day ty.. All

JFK trips average \$46, 4× Midtown's \$11



Late-night riders skip the tip most: about 8% at 3-5 am



1 • Airports cost 4× more

JFK trips average \$46, LaGuardia \$31, Midtown \$11.

2 • Late night = least tipping

About 8% of card trips at 3-5 am leave no tip, vs about 3% in the day.

3 • Time drives volume, place drives price

The busiest hours are short Manhattan trips; the expensive trips start at airports.

Where you get in matters more than when

Location → fare

Airport pickups are high-fare 93–96% of the time; Midtown only 14%.

Time → demand

Time shapes how many trips happen far more than how much they cost or tip.

Next

Train the high-fare classifier, test tip-of-total, add weather and taxi zones.

EVIDENCE OF CONTRIBUTION

Who did what, with the files and commits to prove it

Member	Role	Files in the repo	Commits
Shashidhar Kumar Chintagunta	Data lead	01_cleaning.ipynb, challenges.md, PROJECT_QUESTION.md, 5 cleaning charts, README	1
Sai Shivananda Gunda	EDA lead A	eda_part1.ipynb, eda1–3.png, TableauEDA1.twb, eda_part1_insights.md	1
Geethaamruth V. Tirumalasetty	EDA lead B + ML	03_eda_part2.ipynb, 3 CSVs, visuals 4–6 (.png + .twbx)	1
Vamsi Krishna Bethapudi	Dashboard + slides	NYC_Taxi_Dashboard.twbx (2 new views), dashboard_insights, slides, README	3

Repo: github.com/vamsikrishnabethapudi02-web/nyc-taxi-project · Dashboard: NYC_Taxi_Dashboard.twbx (Tableau)